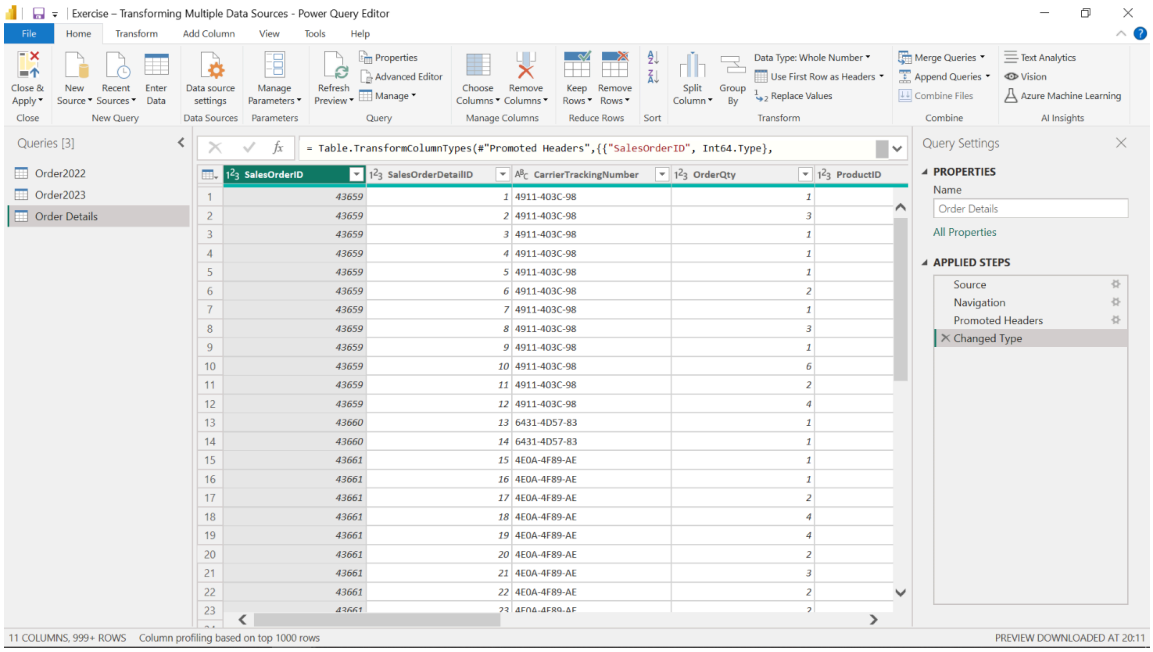
Introduction

Worked with three separate sales datasets in Power Query to clean, profile, and combine them into a single unified table. The task covered end-to-end data preparation from detecting anomalies to merging sources, ready for analysis.

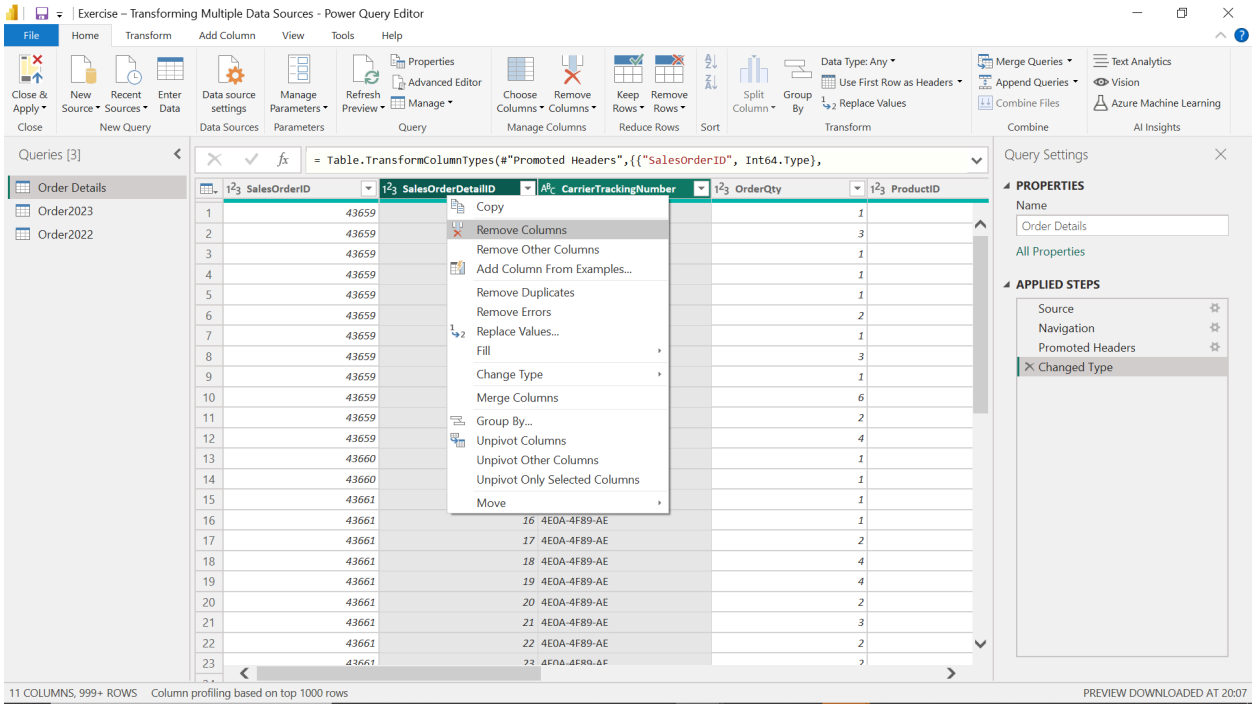
Step 1: Setup the project

Created a new Power BI project and imported three Excel files: Order2022, Order2023, and OrderDetails: using the Get Data feature to open them in Power Query Editor.



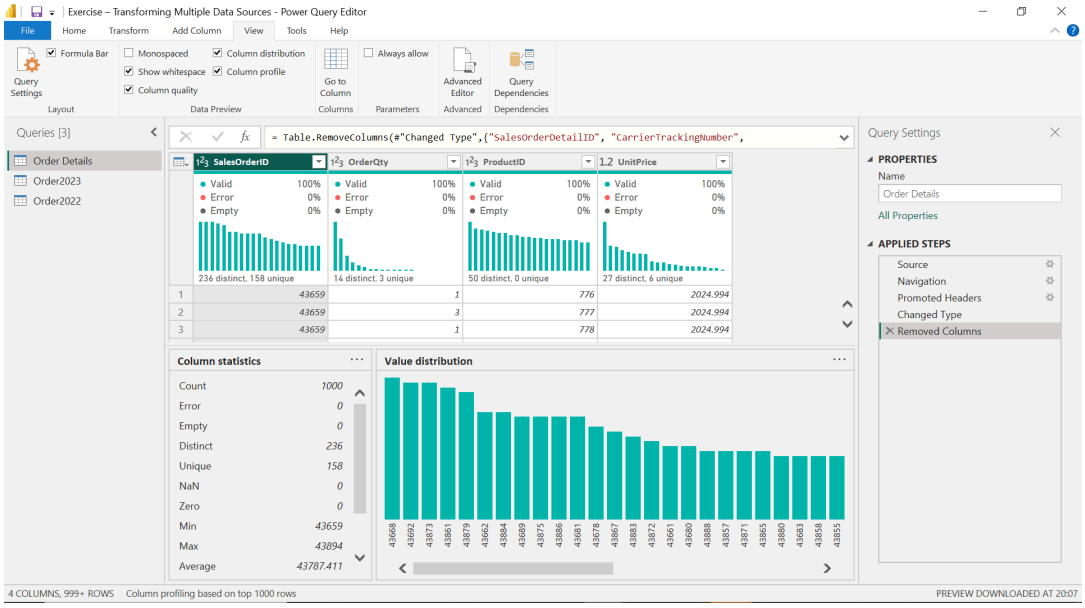
Step 2: Choose columns from OrderDetails

Trimmed the OrderDetails query down to only the four relevant columns: SalesOrderID, ProductID, OrderQty, and UnitPrice.



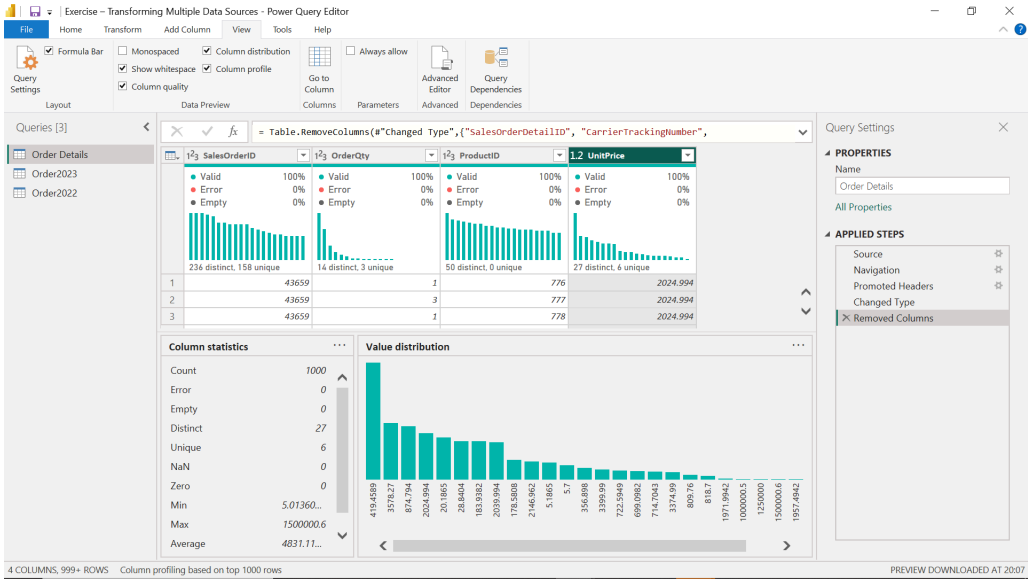
Step 3: Profile data in OrderDetails

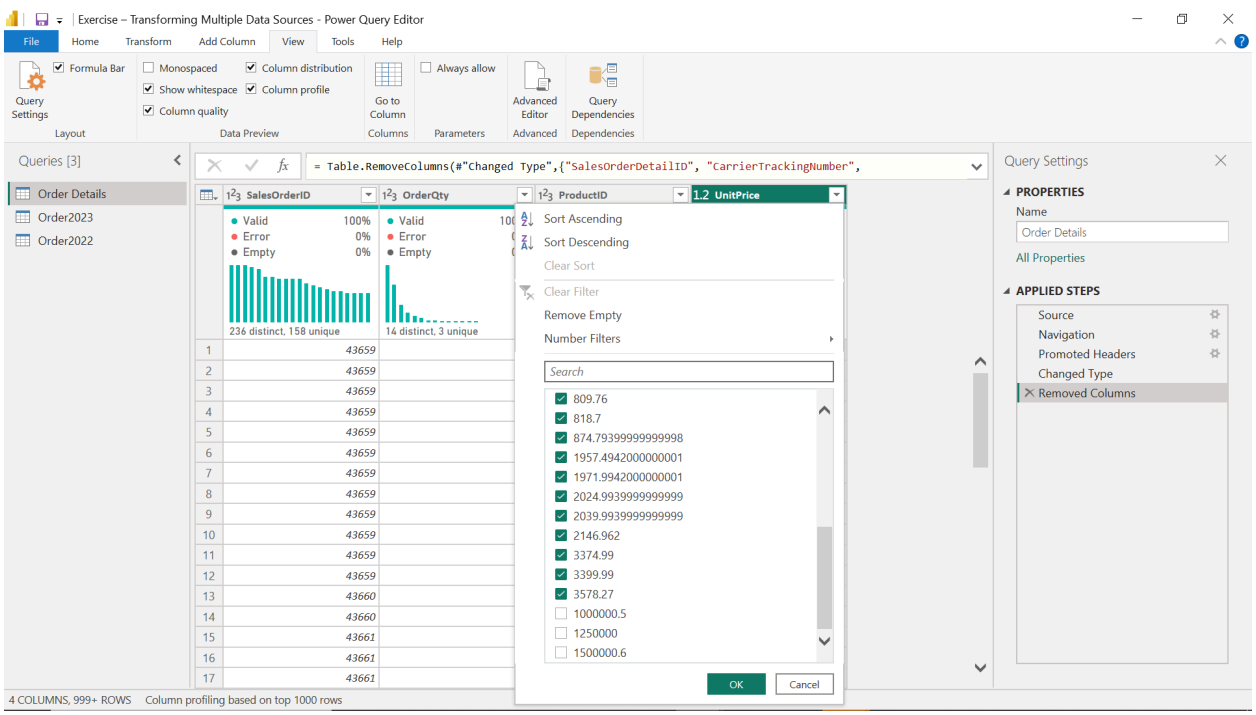
Profiled the OrderDetails data using Column Distribution, Column Quality, and Column Profile to examine valid, error, empty, distinct, and unique values across all columns.



Step 4: Detect potential anomalies in the price

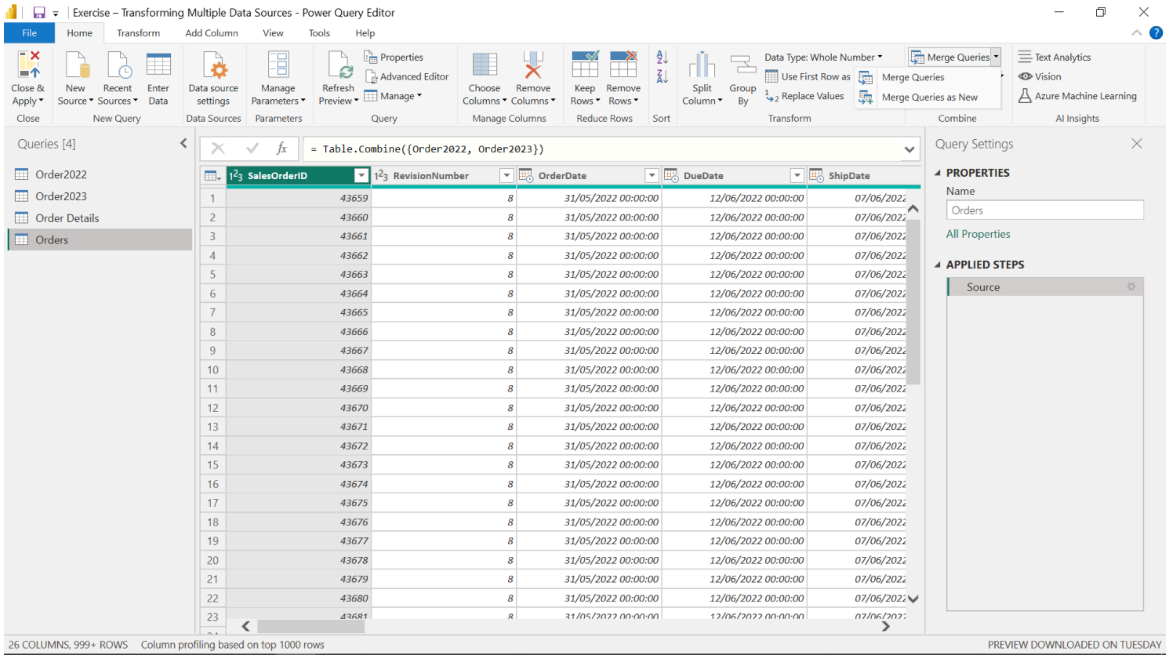
Analysed the UnitPrice column to identify min, max, and mean values, then filtered it to detect and remove three outlier rows flagged as likely data entry errors.





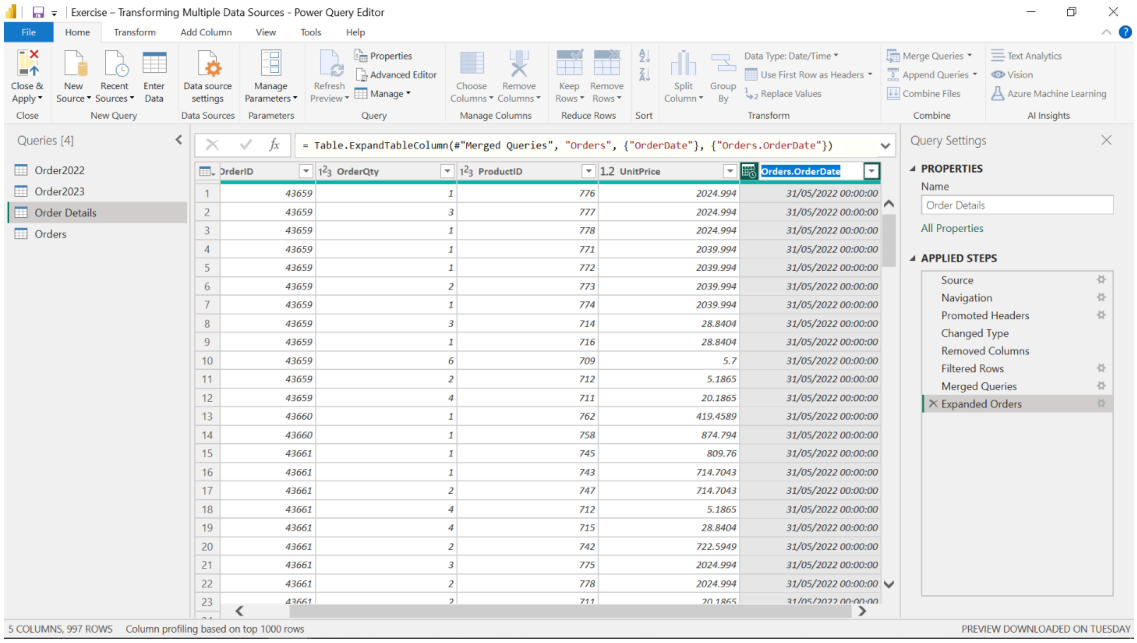
Step 6: Append Queries

Appended Order2022 and Order2023 into a single combined query using Append Queries as New, verified the result, and renamed the query to Orders.

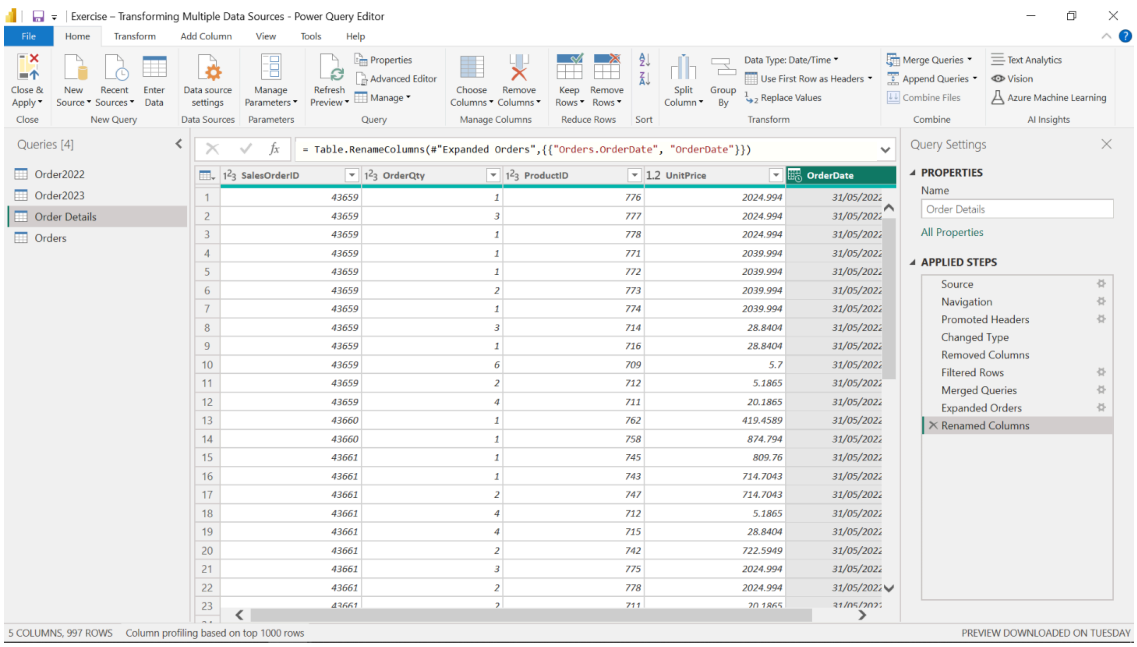


Step 7: Merge Queries:

Merged the OrderDetails and Orders queries on the shared SalesOrderID column using an Inner Join, pulled in the OrderDate column from Orders, and renamed it for clarity.



Completed exercise



Conclusion

Completed a full data preparation workflow in Power Query: cleaning, profiling, anomaly detection, appending, and merging, producing a single reliable dataset ready for analysis.